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NEDUET

CIS- 3rd Year, (Section A, B & C)

EF-305 Engineering Economics and Management

Midterm Exam

Total marks: 20, Time: 1.5 hrs.

Date: 02/April/2026

Instructions: Attempt all the questions. Must write your name, roll number and section on the question paper. Solve Question 7 (Q7) on question paper and submit it with the answer script.

Q1. During the 1990s, technological advancement reduced the cost of computer chips. Discuss how this affected the market for computers? (02) (CLO 02)

Q2. "Rise in price of a commodity is always an effective strategy of producers to increase revenue", Defend or Reject the above statement with appropriate reasoning. (02) (CLO 02)

Q3. John is thinking about opening a café. He estimates that it would cost \$500,000 per year to rent the premises, buy the equipment to make hot drinks and snacks and to buy in the ingredients. In addition, he would have to leave his \$50,000 per year job as an accountant.

(CLO 2)

- Identify, what is John's opportunity cost of running the café for a year? (01)
- If John thought he could sell \$510,000 worth of coffee and snacks in a year, should he open the café? Explain your answer. (02)

Q4. If total cost function of a running a digital toys manufacturing business is given as $TC = c(Q) = 10 - 2(Q)$, where, 'TC' is total cost of toys manufacturing in thousand \$ and 'Q' is the units of toys produced. Express on a graph the behaviour of Total Fixed Cost (FC), Total Variable Cost (VC) and Total Cost (TC) with changing production levels. (03) (CLO 02)

Q5. A firm has the following information available to its managers: Fixed costs are \$1500, price = \$8 and the variable costs are \$0.50 per unit. (CLO 02)

- Compute the break-even output ' Q_{BE} ' for this firm? (01)
- Compute the margin of safety if firm's actual production ' Q_A ' is 250 units of output. (01)
- If firm's targeted profit is \$1500. Identify what should be the firm's actual production ' Q_T '? (01)

35
5

Q6. Describe the law of diminishing marginal returns with the help of graph. (05) (CLO 02)

Q7. Write "T" for True, and "F" for False. (05) (CLO 03)

- i. In a sole proprietorship business owners can generate capital funds through issuing stocks. FT X
- ii. In general partnership business all partners have limited liability to pay off business debts. FF X
- iii. Ownership in a partnership business is represented by shares of stock issued. F X
- iv. A corporation has infinite life. T X
- v. In a corporation, directors are the real owners of the business because they take major decisions of the business. F X
- vi. A partnership business is a business organization that has a separate legal entity from its owners. T X
- vii. In a partnership business, partners are the owners and board of directors are the decision makers. FT X
- viii. To set up a partnership business, owners have to meet extensive formal legal requirements. F X
- ix. In a partnership business, taxes are charged twice i.e. once on owner's profit and once on shareholder's dividend. F X
- x. A dividend income is the profit distributed among shareholders after compensating for all the dues. T X

QUESTION #1:

Due to technical advancement, price has reduced, hence the of computer chip which serves ~~for the market for~~ ^{the material for the production of} computers. Hence, the supply of computer market will increase.

1.5
0.2

caupc } Qc

QUESTION #2:

No, this is not always the case. If the goods are necessity goods then the demand will ~~increase~~ remain the same no matter what the price, ^{hence, increasing the price will result in} revenue increased. But, if the goods are luxury goods, the demand may decrease with the increase in revenue. Hence, not always better revenue. Also, in the case of necessity goods, the competitor may introduce a substitute at low cost, the demand shifts, no increase in revenue.

0.2
0.2

QUESTION #3:

a) The opportunity cost is \$50,000 per year.

As he has to leave the job of accountant that ~~costs~~ is \$50,000 per year.

b) The Implicit Cost = \$500,000

The Explicit Cost = \$50,000

The expected revenue = \$510,000.

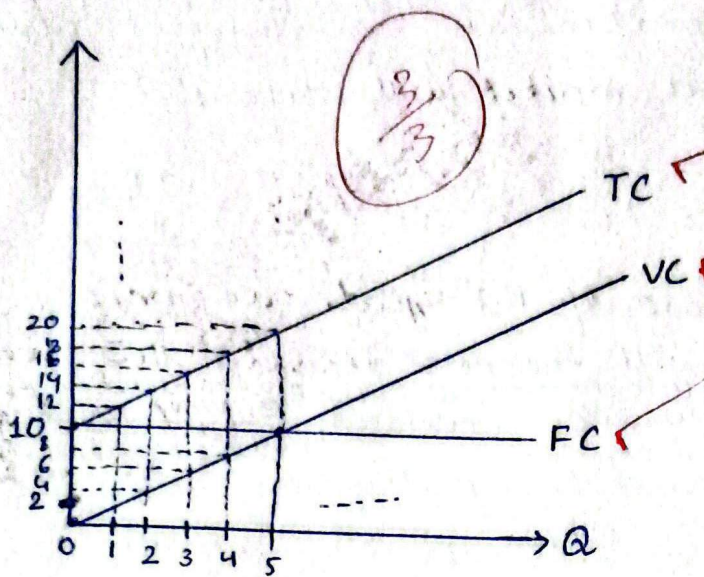
Accounting Revenue = \$510,000 - \$500,000
= \$10,000

Economic Revenue = \$10,000 - \$50,000
= -\$40,000

Since, Economic Revenue < 0 , there is no profit but loss in opening a cafe. Hence, we should not open the cafe.

0.3
0.3

QUESTION #4:



$$VC = 2(Q)$$

$$TC = 10 + 2(Q)$$

$$\text{at } Q=0$$

$$VC=0, TC=10$$

$$\text{at } Q=1$$

$$VC=2, TC=12$$

$$\text{at } Q=2$$

$$VC=4, TC=14$$

$$\text{at } Q=3$$

$$VC=6, TC=16$$

$$\text{at } Q=4$$

$$VC=8, TC=18$$

and so on.

QUESTION #5

$$FC = \$1500$$

$$\text{price} = \$8 \text{ (per unit)}$$

$$VC = \$0.50 \text{ (per unit)}$$

$$\begin{aligned} \text{a) per unit contribution} &= \text{price} - VC \\ &= \$8 - \$0.5 \\ &= \$7.5 \end{aligned}$$

$$Q_{BE} = \frac{FC}{\text{per unit contribution}} = \frac{1500}{7.5} = \boxed{200 \text{ units}}$$

$$\text{b) margin of safety when } Q_A = 250 \text{ units.}$$

$$\begin{aligned} \text{mos} &= (\text{units sold} - \text{break even output}) \times \text{per unit profit} \\ &= (250 - 200)(7.5) \end{aligned}$$

$$\boxed{\text{mos} = \$375}$$